

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Midterm Examination of Semester 2

Year: 2016

Subject: 341152 Electric Circuits II

Section: 5-6

Date: 10 March 2017

Time: 15.00-17.00

Name: _____ ID: _____ Field of Study: _____

Instructions:

1. The examination has 12 pages (including this page), 3 parts (59 questions) and a total score of 75 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

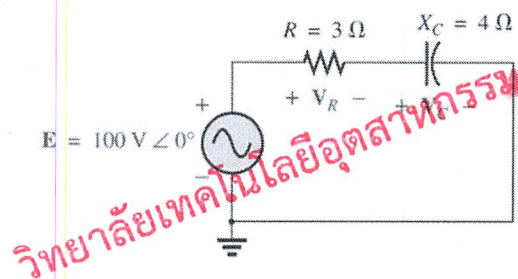
**Cheating in the exam is considered an extremely serious offence
which will result in expulsion from the University.**

Part 1 (25 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Capacitance is directly proportional to the area of the plates.
2. When charging capacitor, the current of capacitor decreases exponentially.
3. The steady-state level of the capacitor current can be found by substituting the open-circuit equivalent.
4. In RC discharging network, the capacitor will be fully discharged after 5τ .
5. The unit of τ is in degree.
6. When charging inductor, the current of inductor increases exponentially.
7. When closing switch of RL DC circuit, electric current cannot flow in right at the beginning.
8. The time constant for an inductor circuit equals its resistance divided by its inductance.
9. An ideal inductor looks like a short circuit to dc current.
10. The current across a capacitor lags the voltage by 90° .
11. Inductive reactance decreases inversely to frequency.
12. Power factor of resistive circuit is equal to zero.
13. For a purely resistive element, the voltage and the current through the element are in phase.
14. In the transient circuit, the time response after 5τ is the steady state area.

15. Consider total power equation $P_T = EI \cos \theta_T$, the θ_T denotes the phase different between E and I .
16. The magnitude of polar form is in effective or RMS value.
17. Resistance value depends on frequency.
18. At very high frequency, the capacitive reactance acts as a short circuit.
19. The unit of capacitive reactance is in F (farad).
20. The unit of average power is in Watt.
21. The unit of power factor $\cos(\theta)$ is in degree.
22. From circuit below, the voltage divider rule can be applied to calculate V_C .



23. The magnitude of the voltage across any one element can be greater than the voltage from source.
24. Given $Z_1 = 5\Omega$ and $Z_2 = j5\Omega$. If Z_1 and Z_2 is connected in the series configuration. Then, $Z_T = j10\Omega$.
25. In parallel network, total admittance is the sum of the individual admittance.

Part 2 (30 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Find the value of capacitor of a small capacitor below.



FIGURE 1

- (a) 200 nF $\pm$ 5% (b) 200 nF $\pm$ 10% (c) 200 pF $\pm$ 5% (d) 200 pF $\pm$ 10%

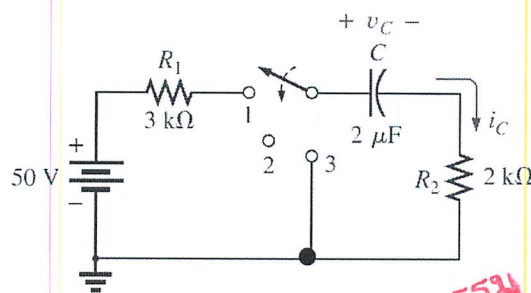


FIGURE 2

2. What is the time constant of the circuit from figure 2?
- (a) 10 ms (b) 4 ms (c) 10 s (d) 6 μ s
3. From figure 2, what is the capacitor voltage at charging state?
- (a) $50e^{t/\tau}$ [V] (b) $50e^{-t/\tau}$ [V] (c) $50(1 - e^{t/\tau})$ [V] (d) $50(1 - e^{-t/\tau})$ [V]
4. From figure 2, what is the capacitor current at charging state?
- (a) $10e^{t/\tau}$ [mA] (b) $10e^{-t/\tau}$ [mA] (c) $5e^{t/\tau}$ [mA] (d) $5e^{-t/\tau}$ [mA]
5. From figure 2, what is the capacitor voltage at discharging state?
- (a) $50e^{t/\tau}$ [V] (b) $50e^{-t/\tau}$ [V] (c) $50(1 - e^{t/\tau})$ [V] (d) $50(1 - e^{-t/\tau})$ [V]
6. From figure 2, what is the capacitor current at discharging state?
- (a) $10e^{t/\tau}$ [mA] (b) $10e^{-t/\tau}$ [mA] (c) $-10e^{t/\tau}$ [mA] (d) $-10e^{-t/\tau}$ [mA]

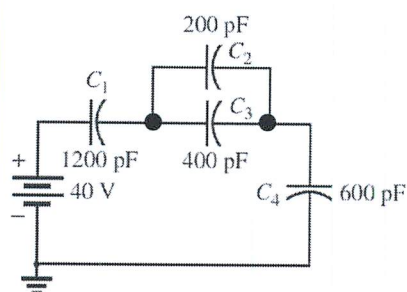


FIGURE 3

7. From figure 3, find the voltage across the capacitor C_1 .
 (a) 30 V (b) 16 V (c) 12 V (d) 8 V
8. From figure 3, find the voltage across the capacitor C_2 .
 (a) 30 V (b) 16 V (c) 12 V (d) 8 V
9. From figure 3, find the energy store in capacitor C_4 .
 (a) 76.8 nJ (b) 76.8 pJ (c) 480 nJ (d) 480 pJ
10. From figure 3, find the energy store in capacitor C_3 .
 (a) 12.8 nJ (b) 12.8 pJ (c) 51.2 nJ (d) 51.2 pJ
11. Find the capacitance of a parallel plate capacitor if the area of each plate is 0.075 m^2 and the distance between the plates is 1.77 mm. The dielectric is paraffin paper ($\epsilon_r = 2.5$) where $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$.
 (a) 0.522 pF (b) 522.45 pF (c) 0.937 pF (d) 938.05 pF
12. What is the unit of inductance?
 (a) Ohm (b) Farad (c) Tesla (d) Henry

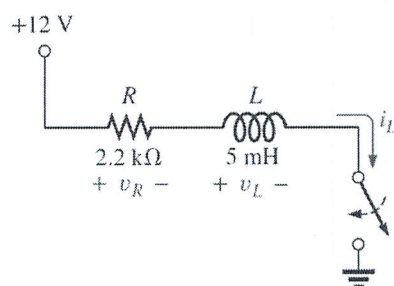


FIGURE 4

19. From Figure 6, what is the time domain equation of voltage (v) and current (i) ?

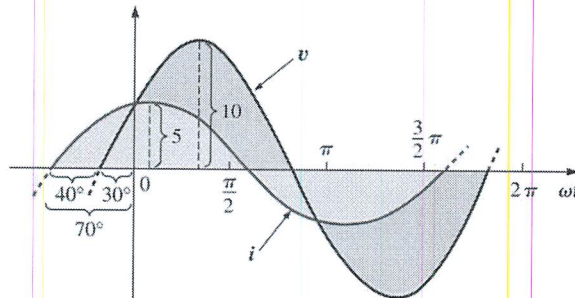


FIGURE 6

(a) $v = 10 \sin(\omega t + 30^\circ)$, $i = 5 \sin(\omega t + 70^\circ)$

(b) $v = 10 \sin(\omega t + 30^\circ)$, $i = 5 \sin(\omega t + 40^\circ)$

(c) $v = 10 \sin(\omega t + 40^\circ)$, $i = 5 \sin(\omega t + 70^\circ)$

(d) $v = 10 \sin(\omega t + 0^\circ)$, $i = 5 \sin(\omega t + 30^\circ)$

20. From Figure 7, what is the phasor form of voltage (v)?

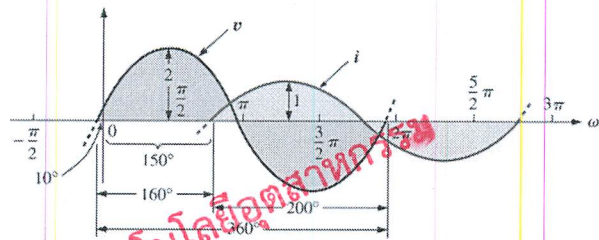


FIGURE 7

(a) $V = \frac{2}{\sqrt{2}} \angle 150^\circ$ (b) $V = \frac{2}{\sqrt{2}} \angle 10^\circ$ (c) $V = \frac{1}{\sqrt{2}} \angle 150^\circ$ (d) $V = \frac{1}{\sqrt{2}} \angle 10^\circ$

21. From question 20, what is the phasor form of current (i)?

(a) $I = \frac{2}{\sqrt{2}} \angle 150^\circ$ (b) $I = \frac{2}{\sqrt{2}} \angle 10^\circ$ (c) $I = \frac{1}{\sqrt{2}} \angle -150^\circ$ (d) $I = \frac{1}{\sqrt{2}} \angle -10^\circ$

22. If 50 V potential difference exists on a parallel-plate capacitor, find the capacitance value of total charge of 1,000 μC .

(a) 42 μF (b) 73 μF (c) 20 μF (d) 28 μF

23. Find the capacitance of a parallel-plate air dielectric capacitor if the area of each of its plates is 0.1 m^2 and the distance between the plates is 2.0 mm.

(a) 1.44 nF (b) 0.62 nF (c) 0.44 nF (d) 0.14 nF

24. What is the inductive reactance at 400 Hz of a 1 mH inductor?

(a) 12 Ω (b) 5.0 Ω (c) 2.5 Ω (d) 0.2 Ω

25. What is the total impedance (Z_T) of the circuit in Figure 8?

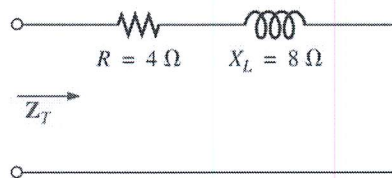


FIGURE 8

- (a) 12Ω (b) $j12\Omega$ (c) $4-j8\Omega$ (d) $4+j8\Omega$

26. At what frequency does a $20\ \mu\text{F}$ capacitor have a reactance of 100Ω ?

- (a) 79.5 kHz (b) 1.59 kHz (c) 79.5 Hz (d) 1.59 MHz

27. The voltage across a 100 mH coil is $v = 50 \sin 50t$. Which of these expressions describes the current?

- (a) $1000 \sin(50t - 90^\circ)$ (b) $10 \sin(50t + 90^\circ)$
(c) $10 \sin 50t$ (d) $10 \sin(50t - 90^\circ)$

28. Which one of the following values is equivalent to $1 \angle 180^\circ$?

- (a) $-1 \angle 0^\circ$ (b) $-1 \angle 90^\circ$ (c) $1 \angle 360^\circ$ (d) $1 \angle -90^\circ$

29. From circuit in Figure 9, find current of inductor (I_L).

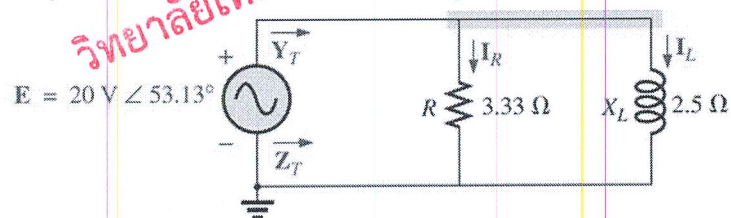


FIGURE 9

- (a) $5 \angle -15^\circ \text{ A}$ (b) $10 \angle 15^\circ \text{ A}$
(c) $8 \angle -36.87^\circ \text{ A}$ (d) $10 \angle -105^\circ \text{ A}$

30. From question 29, find total current of the circuit (I_T)

- (a) $1 \angle -15^\circ \text{ A}$ (b) $10 \angle 0^\circ \text{ A}$
(c) $8 \angle 15^\circ \text{ A}$ (d) $10 \angle -5^\circ \text{ A}$

Part 3 (20 points) Calculation

Instruction: Show the mathematical expression and answer of following problems.

- Find the total ADMITTANCE (Y_T) and IMPEDANCE (Z_T) of the circuit of Figure 10.

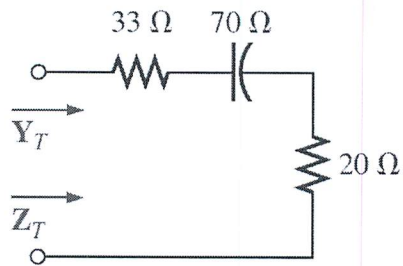


Figure 10

- Determine x and y if $(5x+j10)(2-jy)=90-j70$.

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3. From the circuit in Figure 11, find amplitude of capacitor voltage V_c and current I_c from 1τ to 6τ of the charging state and time constant. Given $E=10\text{ V}$, $R=1\text{ k}\Omega$ and $C=1\mu\text{F}$. Finally, draw the graph of V_c and current I_c time responses.

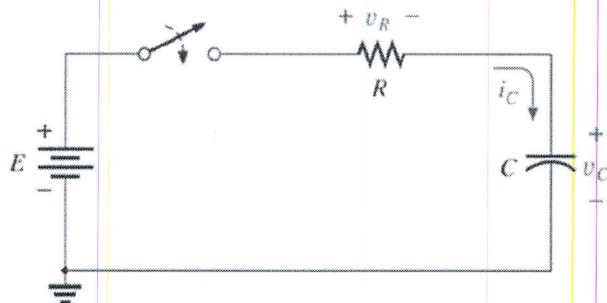


Figure 11

$\tau =$

V_c and I_c at $1\tau =$

V_c and I_c at $2\tau =$

V_c and I_c at $3\tau =$

V_c and I_c at $4\tau =$

V_c and I_c at $5\tau =$

V_c and I_c at $6\tau =$

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4. In the circuit for Figure 12, find V_a and V_b . Given source $e = 141.4214 \sin(\omega t + 30^\circ)$ V.

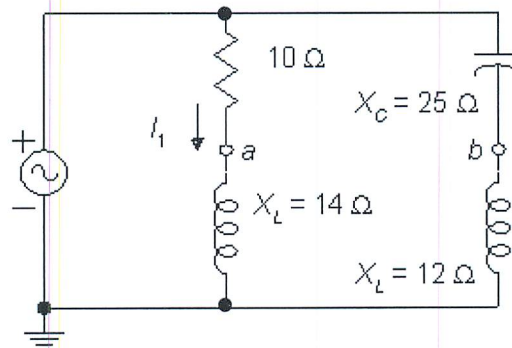
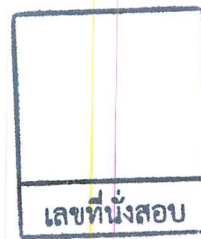


Figure 12

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Answer Sheet



Name: _____ ID: _____

Subject: 341152 Electric Circuits II

Part 1

	True	False
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Part 2

	a	b	c	d
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